Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass

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Abstract

We present Fast3R, a feed-forward transformer architecture for multi-view 3D reconstruction that processes 1000+ unordered, unposed images in a single forward pass. Existing methods such as DUSt3R [28] operate on image pairs and require $O(N^2)$ forward passes followed by costly global alignment, limiting their scalability to large image collections. Fast3R eliminates this bottleneck by encoding all views jointly with a shared Vision Transformer encoder and fusing information across all views simultaneously through an all-to-all attention decoder. To generalize from a modest training window of 20 views to over 1000 views at test time, we introduce image-index sinusoidal positional embeddings combined with position interpolation. A dual-head DPT architecture predicts both local and global pointmaps with per-point confidence, enabling direct 3D reconstruction without any post-hoc optimization. Fast3R achieves state-of-the-art camera pose accuracy on standard benchmarks while running at 251 FPS on multi-view tasks and reconstructing 1000-image scenes in a single pass where pairwise methods exhaust memory.

1. Introduction

Reconstructing 3D structure from multiple photographs is a foundational problem in computer vision, with applications spanning autonomous navigation, augmented reality, robotics, and cultural heritage preservation. Classical structure-from-motion (SfM) pipelines [19, 20] have long served as the workhorse for this task, chaining together feature extraction, pairwise matching, geometric verification, and bundle adjustment into a multi-stage pipeline. While effective, these systems are brittle to textureless regions, repetitive structures, and wide baselines, and their sequential nature limits throughput.

Recent learning-based approaches have made significant strides toward more robust reconstruction. DUSt3R [28] demonstrated that a transformer trained to directly regress dense 3D pointmaps from image pairs can bypass explicit feature matching and triangulation entirely. Its successor MASt3R [9] further grounds the representation with local feature matching, while Spann3R [24] and MonST3R [33] extend the paradigm to sequential and dynamic settings, respectively. However, all of these methods fundamentally operate on *image pairs*: reconstructing N views requires $O(N^2)$ forward passes, followed by a global alignment step that jointly optimizes all pairwise predictions into a coherent coordinate frame. This quadratic cost becomes prohibitive for large-scale scenes; DUSt3R, for example, runs out of memory beyond 48 images on a single GPU.

We propose Fast3R, a transformer architecture that reconstructs an arbitrary number of unordered, unposed images in *one forward pass*. The key insight is that all-to-all attention across views, when equipped with appropriate positional encodings, can directly produce globally consistent 3D pointmaps without any pairwise decomposition or subsequent alignment. Fast3R uses a ViT-Large encoder [8] initialized from CroCo [29] to independently embed each image, then feeds all token sequences into a shared decoder with cross-view all-to-all attention accelerated by FlashAttention [5, 6]. A pair of DPT [14] regression heads decode the fused representations into per-view global and local pointmaps along with confidence scores.

A critical design challenge is enabling the model to generalize to far more views at inference than it sees during training. We address this through image-index sinusoidal positional embeddings: each view receives a positional code based on its index in the input sequence, and we apply position interpolation [4] to extend from the 20-view training regime to 1000+ views at test time with no fine-tuning. During training, random index sampling ensures the model encounters diverse positional configurations, promoting smooth interpolation at arbitrary view counts.

Our contributions are as follows:

- We introduce Fast3R, the first single-forward-pass multi-view 3D reconstruction model that directly predicts globally consistent pointmaps from an arbitrary number of unposed images, eliminating the need for pairwise processing and global alignment.
- We propose image-index positional embeddings with position interpolation, enabling the model to train on 20 views and generalize to 1000+ views at test time without architectural changes or fine-tuning.
- We design a dual-head architecture with local and global pointmap predictions supervised by confidence-weighted regression losses, allowing the model to leverage com-

plementary geometric cues for improved reconstruction fidelity.

- We demonstrate state-of-the-art pose estimation accuracy on CO3Dv2 [15] and RealEstate10K [35] benchmarks while achieving orders-of-magnitude speedups over existing methods—processing 32 views in 0.5 seconds compared to 129 seconds for DUSt3R, and scaling to 1000 views where pairwise methods fail.

2. Related Work

Classical Structure from Motion. Traditional SfM systems [19] decompose reconstruction into sequential stages: feature detection, pairwise matching, geometric verification, and bundle adjustment. Visual SLAM methods such as ORB-SLAM2 [13] operate in an online setting with similar algorithmic building blocks. While mature and widely deployed, these pipelines are sensitive to feature quality and require careful engineering to handle degenerate configurations. Multi-view stereo methods [20, 31] extend reconstruction to dense geometry but typically require known camera poses as input.

Learning-Based Matching and Pose Estimation. Learned feature detectors and matchers such as SuperPoint [7] and SuperGlue [17] improve robustness over classical hand-crafted features, and PixSfM [11] refines keypoints using dense features for sub-pixel accuracy. End-to-end pose regression methods including PoseDiffusion [26], RelPose [34], and RelPose++ [10] bypass explicit matching by directly predicting relative or absolute camera poses from image features. These methods improve over classical pipelines in challenging conditions but remain limited to sparse pose outputs without dense geometry.

Learning-Based Multi-View Reconstruction. Recent methods have sought to modernize the entire SfM pipeline with differentiable or learned components. VGGSfM [25] trains a transformer to produce camera parameters and dense 3D tracks jointly from image collections, achieving strong results on standard benchmarks. FlowMap [22] recovers camera poses, intrinsics, and depth by differentiating through optical flow in a per-scene gradient descent, bypassing hand-crafted feature pipelines. While these methods improve robustness over classical approaches, they still entail per-scene optimization or iterative refinement at test time, preventing true feed-forward throughput. Fast3R eliminates this bottleneck by amortizing all reconstruction computation into a single inference pass.

Feed-Forward 3D Reconstruction. DUSt3R [28] pioneered a transformer-based approach that directly regresses dense 3D pointmaps from image pairs, training on large-scale multi-view data to learn geometric priors. MASt3R [9] extends DUSt3R with local feature heads for improved matching, while Spann3R [24] adapts it to ordered video sequences and MonST3R [33] handles dynamic scenes. FrozenRecon [30] leverages frozen pretrained features for monocular reconstruction. All pairwise methods share a fundamental scalability limitation: $O(N^2)$ inference cost and the need for global alignment to fuse pairwise predictions. Fast3R removes both constraints by processing all views jointly in a single pass.

Scalable Transformers. Applying attention over hundreds of views demands computational efficiency. FlashAttention [5, 6] provides IO-aware exact attention with reduced memory footprint, making all-to-all attention over long token sequences practical. Position interpolation [4], originally proposed for extending context length in large language models, enables transformers to generalize beyond their training sequence lengths. We adapt this technique to the image-index dimension, allowing Fast3R to scale from 20 training views to over 1000 at inference with minimal degradation, and combine it with RoPE [23] spatial encodings in the encoder for within-image positional awareness.

3. Method

3.1. Overview

Given N unordered images $\{I_1, \ldots, I_N\}$ without known camera parameters, Fast3R produces dense 3D pointmaps and confidence maps for every view in a single forward pass. Unlike pairwise methods such as DUSt3R [28] and MASt3R [9], which require $O(N^2)$ forward passes followed by global alignment, Fast3R processes all views jointly with $O(1)$ network evaluations, yielding dramatic speedups at scale.

3.2. Image Encoder

We adopt a Vision Transformer (ViT-L) [8] as the image encoder, initialized from CroCo [29] pre-trained weights via DUSt3R [28]. The encoder uses patch size 16, embedding dimension 1024, 16 attention heads, and 24 transformer layers. We employ Rotary Position Embeddings (RoPE2D) [23] to encode spatial positions within each image, enabling the encoder to handle multiple aspect ratios through a ManyAR patch embedding scheme. FlashAttention [5, 6] is used throughout for memory-efficient computation. Each image is encoded independently, producing a set of patch tokens per view.

3.3. Fusion Decoder

The core of Fast3R is a fusion decoder that enables all-to-all attention across every patch from every view simulta-

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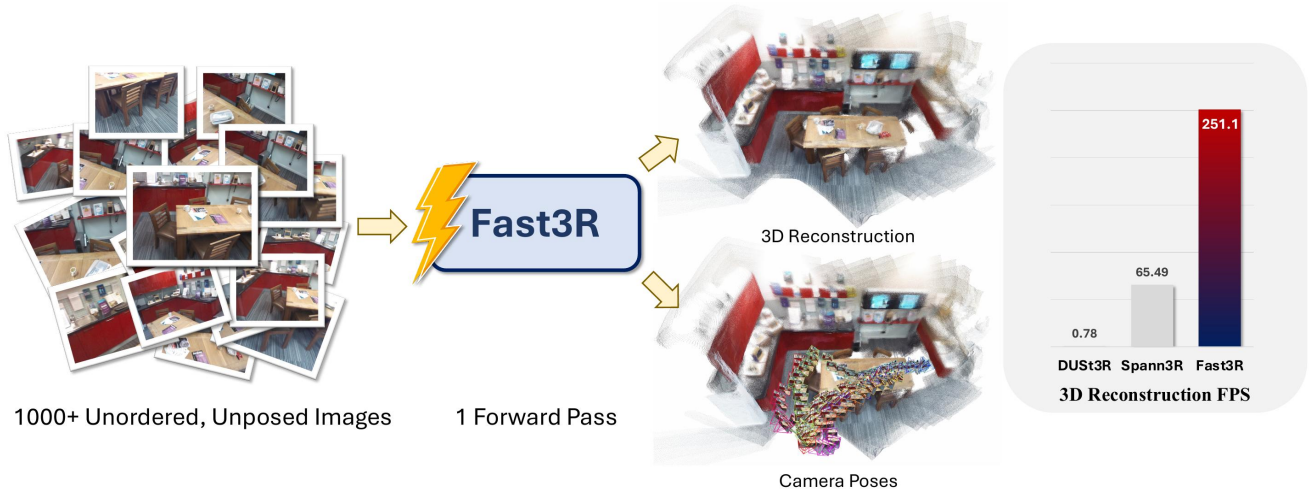

Figure 1. **Fast3R reconstructs 1000+ images in a single forward pass.** Given a large unordered, unposed image collection, Fast3R produces dense 3D pointmaps for all views simultaneously, without any pairwise matching, global alignment, or iterative optimization. This enables orders-of-magnitude speedups over existing methods while maintaining competitive reconstruction accuracy.

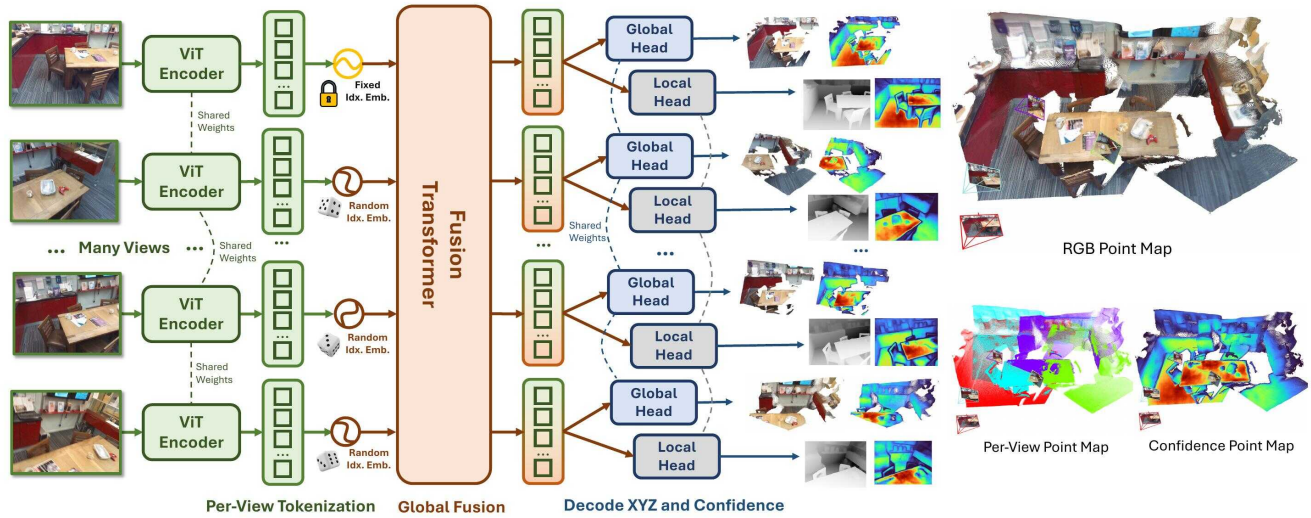

Figure 2. **Architecture of Fast3R.** Given N unordered, unposed images, each image is independently encoded by a shared ViT-L encoder with RoPE2D positional encoding. The resulting patch tokens from all views are concatenated into a single sequence and processed by a fusion decoder with all-to-all attention. Two DPT heads regress per-view 3D pointmaps and confidence scores: a *global* head predicting points in a shared world frame and a *local* head predicting points in each camera's own frame. At inference, local pointmaps are aligned to the global frame for improved reconstruction.

neously. Patch tokens from all N images are concatenated along the sequence dimension into a single sequence and linearly projected from the encoder dimension (1024) to the decoder dimension (768).

To distinguish patches from different images, we add *image-index positional embeddings*: sinusoidal 1D positional encodings computed from each patch's source image index. These embeddings are pre-computed for indices up to 1000, enabling the model to handle an arbitrary number of views.

The decoder consists of 12 transformer blocks with embedding dimension 768, 12 attention heads, and FlashAttention. Crucially, every patch attends to every other patch regardless of source image, allowing the network to jointly reason about multi-view geometry without explicit epipolar constraints or pairwise matching.

3.4. Pointmap Regression Heads

Two DPT [14] heads independently regress dense 3D pointmaps and confidence scores from the decoder features:

- **Global head:** Predicts 3D points in a shared world coordinate frame for all views, establishing cross-view consistency.
- **Local head:** Predicts 3D points in each camera's own coordinate frame, producing more accurate per-view geometry.

Both heads use features from decoder layers at hooks $\{0, \lfloor L/2 \rfloor, \lfloor 3L/4 \rfloor, L\}$ with feature dimension 256 and output dimension 128. At inference, local pointmaps are rigidly aligned to the global frame using confidence-weighted Procrustes alignment, combining the consistency of global predictions with the accuracy of local ones.

3.5. Camera Pose Recovery

Camera poses are recovered directly from the network outputs without any iterative optimization. For each view i, the camera-to-world transformation $T_i \in SE(3)$ is obtained by solving the confidence-weighted Procrustes problem between the local pointmap $\mathbf{P}_i^{\text{loc}}$ and global pointmap $\mathbf{P}_i^{\text{gl}}$:

$$\hat{T}_i = \arg\min_{T \in SE(3)} \sum_p c_p \left\| T\mathbf{P}_i^{\text{loc}}[p] - \mathbf{P}_i^{\text{gl}}[p] \right\|_2^2, \quad (1)$$

where c_p is the per-point confidence predicted by the global head. This closed-form solution requires only a single weighted SVD step, adding negligible overhead to the forward pass. Because both heads are jointly trained with confidence-weighted supervision, the network learns to produce mutually consistent local and global predictions, yielding accurate poses for all views simultaneously without post-hoc refinement.

3.6. Position Interpolation for View Scaling

A key design choice enabling Fast3R to generalize from training on 20 views to inference on 1000+ views is *random image-index sampling*. During training, each image is assigned a random index uniformly sampled from $\{0, \ldots, 999\}$ rather than a sequential index. This strategy, inspired by position interpolation in large language models [4], ensures the model encounters the full range of positional embeddings during training, enabling zero-shot generalization to arbitrary view counts at inference without any positional extrapolation.

3.7. Training

Fast3R is trained with a confidence-weighted regression loss applied to both heads. Specifically, we minimize a normalized $\ell_{2,1}$ loss on predicted pointmaps, weighted by predicted confidence, with an α=0.2 regularization term penalizing low confidence. The normalization uses the average pairwise distance between ground-truth points (`avg_dis`) to ensure scale-invariant supervision. Formally, the per-head loss is:

$$\mathcal{L}_{\text{head}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \frac{c_p}{\bar{c}} \cdot \frac{\|\hat{\mathbf{X}}_p - \mathbf{X}_p\|_2}{\texttt{avg_dis}} - \alpha \log c_p, \quad (2)$$

where $\hat{\mathbf{X}}_p$ is the predicted 3D point, $\mathbf{X}_p$ is the ground-truth point, c_p the predicted confidence, $\bar{c}$ the mean confidence over all points in the batch, and α=0.2 controls confidence regularization strength. The total objective sums the losses from both heads: $\mathcal{L} = \mathcal{L}_{\text{global}} + \mathcal{L}_{\text{local}}$.

We initialize from DUSt3R [28] pre-trained weights for the encoder and downstream heads. Training uses AdamW [12] with learning rate 10^{-4}, β=(0.9, 0.95), and weight decay 0.05, with linear warmup over 10 epochs followed by cosine annealing to 10^{-6} over 500 epochs. We train with 20 views per sample at multiple resolutions (512×384, 512×336, 512×288, 512×256, 512×160) on a mixture of six datasets (80k samples each): CO3Dv2 [15], ScanNet++ [32], ARKitScenes [3], Habitat [18], Blended-MVS, and MegaDepth. We also train a variant, Fast3R-no-outdoor, excluding MegaDepth to focus on indoor and object-centric scenes. Training is performed on 8 A100 GPUs for approximately 100 GPU-hours in total. We apply standard color jitter, random horizontal flipping, and scale augmentation during data loading to improve robustness to photometric variations and diverse scene scales. Gradient norms are clipped to 1.0 to stabilize early training, and mixed-precision (BF16) is used throughout to reduce memory overhead.

4. Experiments

We evaluate Fast3R on camera pose estimation, 3D reconstruction, and system scalability, comparing against state-of-the-art feed-forward and optimization-based methods.

4.1. Camera Pose Estimation

Datasets and metrics. We evaluate on CO3Dv2 [15] and RealEstate10K [35]. Following prior work [26, 28], we report Relative Rotation Accuracy (RRA@τ), Relative Translation Accuracy (RTA@τ), and mean Average Accuracy mAA(30) for both rotation and translation. All methods are evaluated on 10-view subsets; Fast3R does not assume known camera intrinsics.

Results. Table 1 presents the comparison. On CO3Dv2 with 10 views, Fast3R-no-outdoor achieves RRA@15=99.7%, RRA@5=97.4%, and mAA(30)= 82.5, outperforming DUSt3R [28] and MASt3R [9] while being orders of magnitude faster (251.1 FPS vs. 0.78 for DUSt3R and 0.23 for MASt3R).

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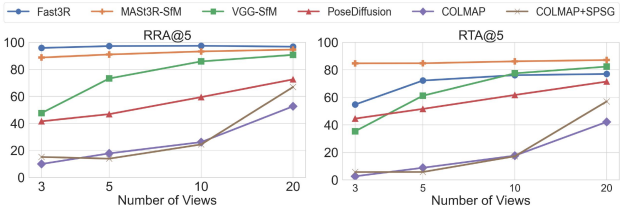

Figure 3. **Pose accuracy improves with more views.** RRA and RTA on CO3Dv2 as a function of the number of input views at test time. Fast3R consistently improves with additional views, demonstrating effective multi-view reasoning.

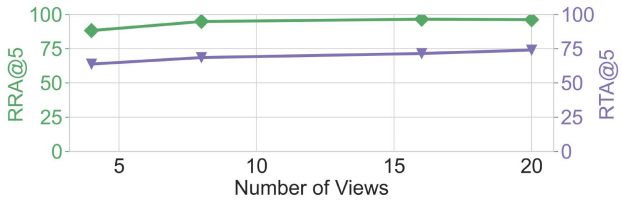

Figure 4. **Effect of training with more views on pose estimation.** Camera pose accuracy on CO3Dv2 improves as the number of views used during training increases, validating the benefit of joint multi-view training with position interpolation.

On RealEstate10K, Fast3R achieves mAA(30)= 72.7, demonstrating competitive performance on in-the-wild scenes. Fast3R substantially outperforms learning-based methods such as PoseDiffusion [26], RelPose [34], and RelPose++ [10], as well as feature-matching pipelines including COLMAP+SuperGlue [7, 16] and PixSfM [11].

Figure 3 shows that Fast3R's pose accuracy consistently improves as the number of input views increases, confirming that the all-to-all attention mechanism effectively leverages additional observations for more accurate geometric reasoning.

4.2. System Scalability

Table 2 compares wall-clock time and peak GPU memory on a single A100 GPU (images at 512×384 resolution) as the number of views increases. Fast3R processes 32 views in 0.509s compared to 129.0s for DUSt3R, a 253× speedup. DUSt3R runs out of memory beyond 48 views, while Fast3R scales to 1000 views in 137.6s using only 63.0 GiB of GPU memory. This scalability is a direct consequence of the single-forward-pass design: Fast3R avoids the $O(N^2)$ pairwise inference and costly global optimization required by DUSt3R.

The memory efficiency of Fast3R stems from its linear scaling in the number of views. With FlashAttention, peak GPU memory grows as $O(N \cdot P)$, where P is the number of patches per image, in contrast to DUSt3R's pairwise scheme which must enumerate all $\binom{N}{2}$ image pairs—an $O(N^2)$ term that quickly dominates GPU memory. This distinction explains why DUSt3R exhausts 80 GiB at merely 48 views while Fast3R comfortably processes 1000 views within the same hardware budget. The practical implication is significant: large-scale applications such as city-block mapping or long-sequence video reconstruction, which require hundreds to thousands of overlapping frames, become tractable with Fast3R on a single GPU without resorting to scene partitioning or out-of-core processing.

4.3. 3D Reconstruction

Scene-level reconstruction. Table 3 reports reconstruction quality on 7-Scenes [21] and NRGBD [2], measured by median accuracy (Acc) and completeness (Comp) scaled by 100×. These datasets contain video trajectories of 500–1500 frames, evaluated using the same frame skip as baselines (skip= 20 for 7-Scenes, skip= 40 for NRGBD). On 7-Scenes, Fast3R achieves Acc= 1.58 and Comp= 0.93, competitive with DUSt3R [28] and better than Spann3R [24]. On NRGBD, Fast3R obtains Acc= 3.40 and the best Comp= 1.01. Crucially, Fast3R achieves these results at 251.1 FPS, over 3× faster than Spann3R and more than 300× faster than DUSt3R.

Object-centric reconstruction. Table 4 shows results on the DTU [1] benchmark (skip= 5 on 49-frame trajectories). Fast3R achieves the best completeness (Comp= 0.857) among compared methods, demonstrating strong reconstruction coverage. The completeness advantage is particularly pronounced because DTU evaluations penalize surface gaps across the full object, rewarding methods that maintain coverage in low-texture and specular regions. Fast3R's joint attention over all views allows confident patches to inform reconstruction of adjacent ambiguous regions, naturally improving coverage without requiring additional post-processing.

Figure 5 illustrates that increasing the number of views during training consistently improves reconstruction quality, validating the benefit of joint multi-view training. Qualitative examples in Figure 6 further confirm that Fast3R produces dense, coherent point clouds across diverse indoor and outdoor scenes without any per-scene optimization.

5. Ablation Studies

We ablate the key design choices of Fast3R: the dual-head architecture with local-to-global alignment, the random image-index positional embedding strategy, and the effect of scaling the number of views.

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Methods	CO3Dv2 [15]					RealEstate10K [35]	FPS
	RRA@15↑	RRA@5↑	RTA@15↑	RTA@5↑	mAA(30)↑	mAA(30)↑	
Colmap+SG [7, 16]	36.1	24.4	27.3	17.2	25.3	45.2	0.056
PixSfM [11]	33.7	26.1	32.9	17.6	30.1	49.4	-
RelPose [34]	57.1	-	-	-	-	-	0.02
PosReg [26]	53.2	-	49.1	-	45.0	-	0.015
PoseDiff [26]	80.5	59.5	79.8	61.7	66.5	48.0	0.015
RelPose++ [10]	(85.5)	-	-	-	-	-	0.02
DUSt3R [27]	96.2	-	86.8	-	76.7	67.7	0.78
MASt3R [9]	94.6	93.2	**91.9**	**86.2**	81.8	**76.4**	0.23
Fast3R-no-outdoor (Ours)	**99.7**	**97.4**	87.1	76.1	**82.5**	-	**251.1**
Fast3R (Ours)	96.2	90.2	81.6	68.2	75.0	72.7	**251.1**

Table 1. **Multi-view pose regression on the CO3D [15] and RealEstate10K [35] datasets**. Parentheses denote methods that do not report results on the 10 views set; we report their best for comparison (8 views). Fast3R does not assume known camera intrinsics.

# Views	Fast3R		DUSt3R	
	Time (s)	Peak GPU Mem (GiB)	Time (s)	Peak GPU Mem (GiB)
2	0.065	3.84	0.092	3.52
8	0.122	6.33	8.386	24.59
32	0.509	13.25	129.0	67.61
48	0.84	20.8	OOM	OOM
320	15.938	41.90	OOM	OOM
800	89.569	55.97	OOM	OOM
1000	137.62	63.01	OOM	OOM
1500	308.85	78.59	OOM	OOM

Table 2. **System performance metrics for different view counts on Fast3R and DUSt3R on a single A100.** Time is measured in seconds (s), and memory is measured in gibibytes (GiB). Each view is 512x384 in resolution. For DUSt3R, at 48 views the N^2 pairwise reconstructions eventually consume all VRAM at its global alignment stage. Note that Fast3R's reported fastest FPS of 251.1 uses 108 views in 224x224 resolution.

Method	FPS	7 scenes [21]		NRGBD [2]	
		Acc↓	Comp↓	Acc↓	Comp↓
F-Recon [30]	<0.1	7.62	2.31	20.59	6.31
DUSt3R† [28]	0.78	**1.23**	0.91	**2.51**	1.03
Spann3R [24]	65.4	1.48	**0.85**	3.15	1.10
Fast3R (Ours)	**251.1**	1.58	0.93	3.40	**1.01**

Table 3. **Quantitative reconstruction results on scene datasets:** The numbers indicate median distance to GT points on 7-Scenes [21] and NRGBD [2] datasets. These datasets contain video trajectories of 500-1500 frames, and we evaluate using the same frame skip as other baselines. For 7-Scenes this is `skip`=20, and NRGBD uses `skip`=40. DUSt3R† indicates using DUSt3R's final weights on 224 × 224 images, to fit within GPU memory. Distances are scaled 100× to remove the leading `0.00`.

5.1. Local vs. Global Pointmap Head

Table 5 compares using only the global pointmap against aligning the local pointmap to the global frame via confidence-weighted Procrustes alignment. The local-to-global alignment consistently reduces reconstruction error on 7-Scenes (Acc improves by 1.97 cm) and NRGBD (Acc improves by 0.46 cm). On DTU, the difference is marginal, suggesting that the local head is most beneficial for scene-level reconstruction where per-view accuracy matters most.

Method	Views	Acc↓	Comp↓
		Med.	Med.
DUSt3R [28]	all/5	**1.159**	0.914
DUSt3R† [28]	all/5	1.297	1.002
Spann3R [24]	all/5	2.268	1.295
Fast3R (Ours)	all/5	1.706	**0.857**

Table 4. **Quantitative results on object-centric DTU [1] dataset.** Using a `skip`=5 on trajectories of 49 frames.

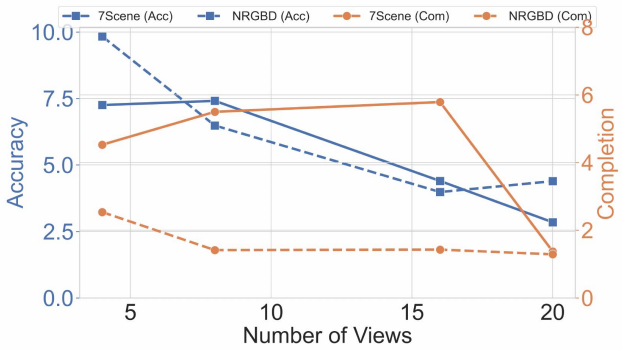

Figure 5. **Effect of training with more views on reconstruction quality.** Increasing the number of views during training improves reconstruction accuracy on 7-Scenes and NRGBD datasets.

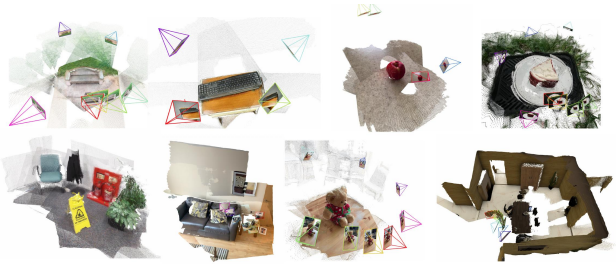

Figure 6. **Qualitative reconstruction examples.** Fast3R produces dense and coherent 3D pointmaps across diverse indoor and outdoor scenes from unordered, unposed image collections.

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Pointmap Type	7-Scenes [21]		NRGBD [2]		DTU	
	Acc↓	Comp↓	Acc↓	Comp↓	Acc↓	Comp↓
local aligned to global	2.84	1.37	4.39	1.28	3.91	1.41
global	4.81	1.64	4.85	1.32	3.88	1.41
Δ	+1.97	+0.27	+0.46	+0.04	-0.03	0.00

Table 5. **Ablation on the effect of local head on 3D reconstruction.** Red/green indicate an increase/decrease in error compared to using the local pointmap aligned to the global pointmap.

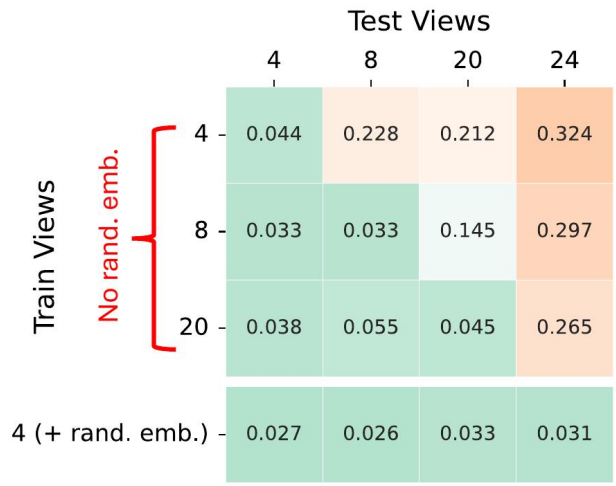

Figure 7. **Effect of random image-index sampling during training.** Without random sampling, performance degrades significantly when the number of test views exceeds the training count of 20. Random sampling over indices $\{0, \ldots, 999\}$ enables smooth generalization via interpolation to 1000+ views.

5.2. Random Image-Index Positional Embedding

Figure 7 demonstrates the importance of random image-index sampling (Section 3.6). When indices are assigned sequentially during training, the model must extrapolate to unseen positional embeddings at test time, causing significant performance degradation beyond the training view count of 20. Random sampling from $\{0, \ldots, 999\}$ enables the model to interpolate rather than extrapolate, maintaining strong performance across a wide range of view counts.

5.3. Scaling with Number of Views

Figure 3 shows that pose accuracy on CO3Dv2 improves monotonically as the number of test views increases. Similarly, Figure 5 demonstrates that training with more views improves downstream reconstruction quality on 7-Scenes and NRGBD. Figure 4 shows consistent improvement in camera pose estimation with training view count. These results confirm that Fast3R's all-to-all attention mechanism effectively leverages additional views, and that the position interpolation strategy enables this scaling without architectural changes or additional fine-tuning. This smooth scalability follows directly from the sinusoidal embedding design: because adjacent image indices yield smoothly varying embeddings, the model reliably interpolates between positions seen during training rather than extrapolating to arbitrary codes. By contrast, a model using learned positional embeddings would require additional fine-tuning whenever the number of test views falls outside the training range, undermining the zero-shot scalability that makes Fast3R practical for large-scale deployments.

6. Conclusion

We presented Fast3R, a feed-forward architecture for multi-view 3D reconstruction that processes 1000+ unposed images in a single forward pass. By concatenating all image patch tokens into one sequence and applying all-to-all transformer attention, Fast3R jointly reasons about multi-view geometry without pairwise inference or global optimization. A random image-index positional embedding strategy enables zero-shot generalization from 20 training views to over 1000 test views through position interpolation. The dual pointmap head design, combining global cross-view consistency with local per-view accuracy, further improves reconstruction quality across scene and object-centric benchmarks.

Fast3R achieves state-of-the-art efficiency with up to 253× speedup over DUSt3R [28] and orders-of-magnitude speedups compared to MASt3R [9], while maintaining competitive accuracy on camera pose estimation and 3D reconstruction. Future work includes integrating fine-grained feature matching for improved correspondence precision, extending to video sequences with temporal priors, and handling dynamic scenes with moving objects. We also envision Fast3R as an efficient initializer for downstream scene representations such as Neural Radiance Fields and 3D Gaussian Splatting, supplying both dense geometry and camera poses without any per-scene optimization overhead.

Limitations. Despite its strong performance and scalability, Fast3R has several limitations. First, the model is trained and evaluated on static scenes; scenes with significant dynamic content or rapid inter-frame motion can degrade pointmap quality. Second, while the decoder's all-to-all attention is linearized in views through FlashAttention, it scales quadratically in the total token count, making ultra-high-resolution inputs memory-intensive at very large view counts; hierarchical or sparse attention variants could mitigate this. Finally, Fast3R does not explicitly recover camera intrinsics; downstream tasks requiring precise focal-length estimates must rely on additional calibration steps.

References

[1] Henrik Aanæs, Rasmus Ramsbøl Jensen, George Vogiatzis, Engin Tola, and Anders Bjorholm Dahl.

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Large-scale data for multiple-view stereopsis. *International Journal of Computer Vision*, 120:153–168, 2016. 5, 6

[2] Dejan Azinović, Ricardo Martin-Brualla, Dan B Goldman, Matthias Nießner, and Justus Thies. Neural rgb-d surface reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6290–6301, 2022. 5, 6, 7

[3] Gilad Baruch, Zhuoyuan Chen, Afshin Dehghan, Tal Dimry, Yuri Feigin, Peter Fu, Thomas Gebauer, Brandon Joffe, Daniel Kurz, Arik Schwartz, and Elad Shulman. Arkitscenes: A diverse real-world dataset for 3d indoor scene understanding using mobile rgb-d data. *arXiv preprint arXiv:2111.08897*, 2021. 4

[4] Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. 1, 2, 4

[5] Tri Dao. FlashAttention-2: Faster attention with better parallelism and work partitioning. In *International Conference on Learning Representations (ICLR)*, 2024. 1, 2

[6] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. 1, 2

[7] Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. Superpoint: Self-supervised interest point detection and description. *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 337–33712, 2017. 2, 5, 6

[8] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021. 1, 2

[9] Vincent Leroy, Yohann Cabon, and Jérôme Revaud. Grounding image matching in 3d with mast3r. *arXiv preprint arXiv:2406.09756*, 2024. 1, 2, 4, 6, 7

[10] Amy Lin, Jason Y Zhang, Deva Ramanan, and Shubham Tulsiani. Relpose++: Recovering 6d poses from sparse-view observations. *arXiv preprint arXiv:2305.04926*, 2023. 2, 5, 6

[11] Philipp Lindenberger, Paul-Edouard Sarlin, Viktor Larsson, and Marc Pollefeys. Pixel-perfect structure-from-motion with featuremetric refinement. *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 5967–5977, 2021. 2, 5, 6

[12] Ilya Loshchilov and Frank Hutter. Fixing weight decay regularization in adam. *CoRR*, abs/1711.05101, 2017. 4

[13] Raúl Mur-Artal and Juan D. Tardós. ORB-SLAM2: an open-source SLAM system for monocular, stereo and RGB-D cameras. *IEEE Transactions on Robotics*, 33(5):1255–1262, 2017. 2

[14] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction. *arXiv preprint arXiv:2103.13413*, 2021. 1, 4

[15] Jeremy Reizenstein, Roman Shapovalov, Philipp Henzler, Luca Sbordone, Patrick Labatut, and David Novotny. Common objects in 3d: Large-scale learning and evaluation of real-life 3d category reconstruction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10901–10911, 2021. 2, 4, 6

[16] Paul-Edouard Sarlin, Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. Superglue: Learning feature matching with graph neural networks. *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4937–4946, 2019. 5, 6

[17] Paul-Edouard Sarlin, Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. SuperGlue: Learning feature matching with graph neural networks. In *CVPR*, 2020. 2

[18] Manolis Savva, Abhishek Kadian, Oleksandr Maksymets, Yili Zhao, Erik Wijmans, Bhavana Jain, Julian Straub, Jia Liu, Vladlen Koltun, Jitendra Malik, Devi Parikh, and Dhruv Batra. Habitat: A platform for embodied AI research. *CoRR*, abs/1904.01201, 2019. 4

[19] Johannes Lutz Schönberger and Jan-Michael Frahm. Structure-from-motion revisited. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. 1, 2

[20] Johannes Lutz Schönberger, Enliang Zheng, Marc Pollefeys, and Jan-Michael Frahm. Pixelwise view selection for unstructured multi-view stereo. In *European Conference on Computer Vision (ECCV)*, 2016. 1, 2

[21] Jamie Shotton, Ben Glocker, Christopher Zach, Shahram Izadi, Antonio Criminisi, and Andrew Fitzgibbon. Scene coordinate regression forests for camera relocalization in rgb-d images. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2930–2937, 2013. 5, 6, 7

[22] Cameron Smith, David Charatan, Ayush Kumar Tewari, and Vincent Sitzmann. Flowmap: High-quality camera poses, intrinsics, and depth via gradient descent. *ArXiv*, abs/2404.15259, 2024. 2

AI-Generated

[23] Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding. *arXiv preprint arXiv:2104.09864*, 2021. 2

[24] Hengyi Wang and Lourdes Agapito. 3d reconstruction with spatial memory. *arXiv preprint arXiv:2408.16061*, 2024. 1, 2, 5, 6

[25] Jianyuan Wang, Nikita Karaev, Christian Rupprecht, and David Novotny. Vggsfm: Visual geometry grounded deep structure from motion. *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21686–21697, 2023. 2

[26] Jianyuan Wang, C. Rupprecht, and David Novotný. Posediffusion: Solving pose estimation via diffusion-aided bundle adjustment. *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9739–9749, 2023. 2, 4, 5, 6

[27] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jérôme Revaud. Dust3r: Geometric 3d vision made easy. *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20697–20709, 2023. 6

[28] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *CVPR*, 2024. 1, 2, 4, 5, 6, 7

[29] Philippe Weinzaepfel, Vincent Leroy, Thomas Lucas, Romain Brégier, Yohann Cabon, Vaibhav Arora, Leonid Antsfeld, Boris Chidlovskii, Gabriela Csurka, and Jérôme Revaud. Croco: Self-supervised pre-training for 3d vision tasks by cross-view completion. *Advances in Neural Information Processing Systems*, 35:3502–3516, 2022. 1, 2

[30] Guangkai Xu, Wei Yin, Hao Chen, Chunhua Shen, Kai Cheng, and Feng Zhao. Frozenrecon: Pose-free 3d scene reconstruction with frozen depth models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9310–9320, 2023. 2, 6

[31] Yao Yao, Zixin Luo, Shiwei Li, Tian Fang, and Long Quan. Mvsnet: Depth inference for unstructured multi-view stereo. *European Conference on Computer Vision (ECCV)*, 2018. 2

[32] Chandan Yeshwanth, Yueh-Cheng Liu, Matthias Nießner, and Angela Dai. Scannet++: A high-fidelity dataset of 3d indoor scenes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 12–22, 2023. 4

[33] Junyi Zhang, Charles Herrmann, Junhwa Hur, Varun Jampani, Trevor Darrell, Forrester Cole, Deqing Sun, and Ming-Hsuan Yang. Monst3r: A simple approach for estimating geometry in the presence of motion. *arXiv preprint arxiv:2410.03825*, 2024. 1, 2

[34] Jason Y. Zhang, Deva Ramanan, and Shubham Tulsiani. Relpose: Predicting probabilistic relative rotation for single objects in the wild. *ArXiv*, abs/2208.05963, 2022. 2, 5, 6

[35] Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyffe, and Noah Snavely. Stereo magnification: learning view synthesis using multiplane images. *ACM Trans. Graph.*, 37(4), 2018. 2, 4, 6

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